

WaveLens Lite — Technical Architecture Report

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Deployment Context: All screenshots are captured from a real, fully operational instance running on local hardware — a laptop paired with a mobile device over Bluetooth earphone. The system is **not yet deployed to a cloud server** — this phase validates the complete end-to-end pipeline on physical hardware before production deployment.

Executive Summary

WaveLens Lite is a production-grade, real-time multilingual voice translation system purpose-built for industrial edge environments — specifically Vietnamese maritime port operations. It captures speech through a connected Bluetooth earphone, routes it through the **Agora Conversational AI Engine (CAI)**, and returns synthesized translated audio and live text subtitles in under 2 seconds. Every session is cryptographically signed and anchored to the Solana blockchain as an immutable compliance audit record.

This document provides an expert, end-to-end architectural analysis from the perspective of a production AI Systems Engineer, detailing every subsystem, its engineering rationale, known failure modes, and implemented mitigations.

1. System-Level Overview

Diagram 0

2. Dual-Channel Real-Time Translation Pipeline

The core of WaveLens is a bidirectional audio/text pipeline over the **Agora SDRTN**. Two parallel channels run simultaneously: the audio channel carries synthesized translated speech, and the data channel carries UTF-8 JSON text chunks for real-time subtitle rendering.

Diagram 1

Engineering Rationale

Constraint	Why It Matters
AEC: true	Prevents the LLM from hearing its own translated output and re-translating it, causing a feedback loop
ANS: true	Da Nang port machinery produces 85-110 dB noise; suppression maintains clean VAD thresholds
AGC: true	Normalizes volume across workers shouting in wind vs. speaking quietly in a confined compartment

Agora SDRTN	Purpose-built real-time relay network with lower jitter than commodity WebRTC in high-packet-loss environments
Dual data channel	Text stream runs parallel to audio — if audio is too noisy, subtitles provide a visual fallback

3. State Machine Lifecycle

This is the most critical engineering layer. A single broken state transition is the root cause of the `INVALID_OPERATION` publish error — the most common failure mode in real-world Agora deployments.

Diagram 2

[!IMPORTANT] **Key fix applied:** `client.publish()` is called **only after** `connectionState === 'CONNECTED'` is confirmed via a Promise-based state listener. The `LiveSession` component is kept **permanently mounted in the DOM** (CSS `opacity: 0` when ready) to prevent React Strict Mode's double-mount from destroying the live RTC session. This eliminated the `INVALID_OPERATION: Can't publish stream, haven't joined yet` error.

4. Mobile Hardware Resilience Architecture

Industrial deployments face OS-level threats invisible in a development environment. Four threat categories are mitigated at the browser layer.

Diagram 3

Microphone Signal Processing Stack

Diagram 4

5. Blockchain Audit Trail — Solana / Metaplex Core

Industrial safety compliance mandates immutable, tamper-proof logging. Storing raw transcripts on-chain is cost-prohibitive. WaveLens uses a **cryptographic hash commitment** model.

Diagram 5

[!NOTE] **Privacy-preserving design:** Only the SHA-256 hash is stored on-chain. The full bilingual transcript remains off-chain (IPFS/Arweave, access-controlled). This satisfies both audit requirements and data sovereignty concerns common in Vietnamese maritime operations.

6. Security Architecture

Diagram 6

[!CAUTION] The App Certificate, Customer Secret, and Solana Private Key are **never** sent to the browser. All sensitive operations run exclusively in Node.js API routes. The

.env.local file is excluded from version control via .gitignore.

7. Technology Stack

Layer	Technology	Rationale
Frontend Framework	Next.js 15 + Turbopack	App Router + Server Components; fast HMR for development
UI Language	TypeScript + React 18	Strict type safety for complex async state machines
RTC SDK	agora-rtc-sdk-ng	Browser-native WebRTC with SDRTN global relay
AI Agent SDK	agora-agents	First-party SDK for full CAI pipeline orchestration
STT	Deepgram Nova-3	Best available Vietnamese ASR; real-time streaming
LLM	Large Language Model	Best cost/latency ratio for translation at under 1 second
TTS	MiniMax speech_2_6_turbo	Sub-200ms synthesis; natural English voice output
Token Auth	agora-token RtcTokenBuilder	Server-side PUBLISHER role tokens; 1-hour expiry
Blockchain	Solana Devnet + Umi + Metaplex Core	Sub-cent transaction cost; soulbound NFT audit model
Styling	CSS + Tailwind	Canvas waveform, glassmorphism UI, mobile-first layout

8. Known Constraints and Engineering Trade-offs

Constraint	Root Cause	Current Mitigation	Future Improvement
0.48s VAD silence threshold	Deepgram/Agora VAD minimum	silence_duration_ms: 480	Implement incremental streaming VAD
End-to-end latency 1.2-2.5s	STT + LLM + TTS in serial	Low-temperature LLM for fast decoding	Streaming LLM output directly to TTS
HFP 16kHz audio quality	Bluetooth specification limit	AEC + ANS + AGC at WebRTC layer	USB-C microphone bypass evaluation
iOS AudioContext suspend	Known Safari regression	visibilitychange resume hook	Persistent Service Worker + WakeLock
No offline fallback	Cloud-only CAI engine	Accepted for MVP scope	Edge-deployed Whisper.cpp for offline STT
Token 1-hour expiry	Agora RTC token TTL	idleTimeout 300s keeps agent alive	Auto-refresh token before expiry window

9. Live Demo Evidence

Note on Deployment Status: The following screenshots are from a fully operational instance of WaveLens Lite running on local hardware. The system is **not yet deployed to a cloud server** — this phase validates the complete hardware-to-cloud pipeline on physical devices before production deployment.

Screenshot 1 — Initial Loading State

The 5-step initialization pipeline: Engine — Microphone — Channel — Agent — RTT

 WaveLens Demo - Initial Loading Screen

Screenshot 2 — Session Start

Channel joined, Agora agent starting, microphone track published to SDRTN

 WaveLens Demo - Session Starting

Screenshot 3 — Live Listening on Mobile

Microphone active, waveform animated in orange, Listening indicator active at bottom

 WaveLens Demo - Live on Mobile Phone

Screenshot 4 — Hardware Field Configuration

Bluetooth earphone worn as intended for real-world use — paired directly to the mobile device, providing clear audio input and synthesized translated speech output during live sessions

 WaveLens Hardware - Bluetooth Earphone Field Configuration

10. Conclusion

WaveLens Lite represents a practical, production-validated approach to a real industrial problem: the communication barrier between Vietnamese port workers and international ship crews. The architecture makes principled engineering trade-offs at every layer.

- **Reliability over features:** The state machine and connection guards prioritize zero-crash operation.
- **Privacy over convenience:** Blockchain audit uses hash-only commitment, not raw transcripts.
- **Resilience over optimization:** VAD thresholds and hardware constraints are tuned for a 90 dB industrial floor, not a quiet office.

The full pipeline — from Bluetooth earphone microphone input, through Agora CAI cloud processing, to synthesized English speech in the earphone — is validated and operational on physical hardware. The remaining steps before port deployment are: cloud hosting on Vercel, an `https://` domain (required for mobile microphone access), and VAD latency fine-tuning for long-form command translation.
